

Quality Report



Generated with Pix4Denterprise version 4.5.6



Important: Click on the different icons for:



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Additional information about the sections



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Summary



Project	Huasca_MAX_V2
Processed	2026-07-07 16:23:15
Camera Model Name(s)	M4E_12.3_5280x3956 (1581F7FVC261F00DN1E3) (RGB)
Average Ground Sampling Distance (GSD)	1.76 cm / 0.69 in
Area Covered	0.066 km ² / 6.5806 ha / 0.03 sq. mi. / 16.2695 acres
Time for Initial Processing (without report)	27m:24s

Quality Check



Images	median of 76571 keypoints per image	
Dataset	379 out of 436 images calibrated (86%), all images enabled, 4 blocks	
Camera Optimization	2.57% relative difference between initial and optimized internal camera parameters	
Matching	median of 11458.5 matches per calibrated image	
Georeferencing	yes, no 3D GCP	

Preview

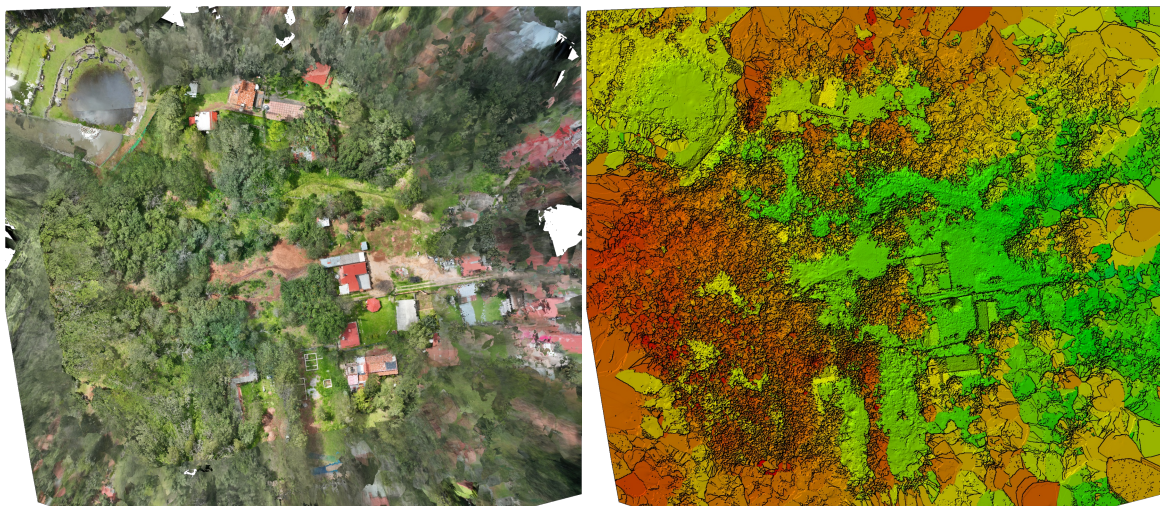


Figure 1: Orthomosaic and the corresponding sparse Digital Surface Model (DSM) before densification.

Calibration Details



Number of Calibrated Images	379 out of 436
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? Initial Image Positions

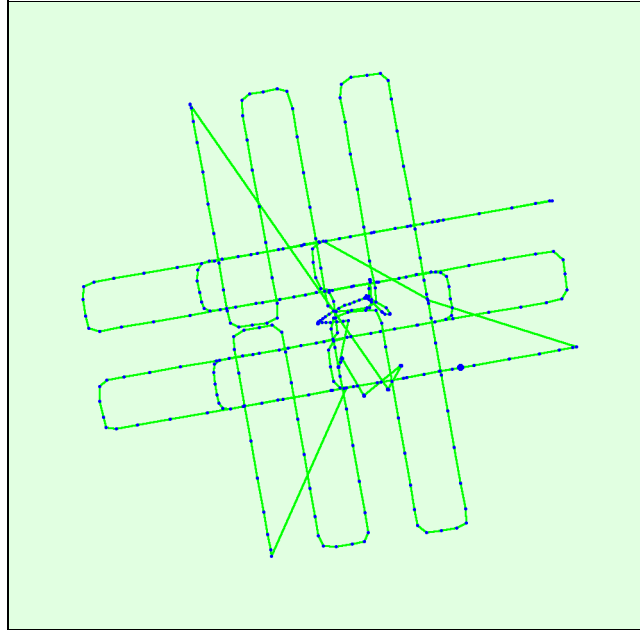
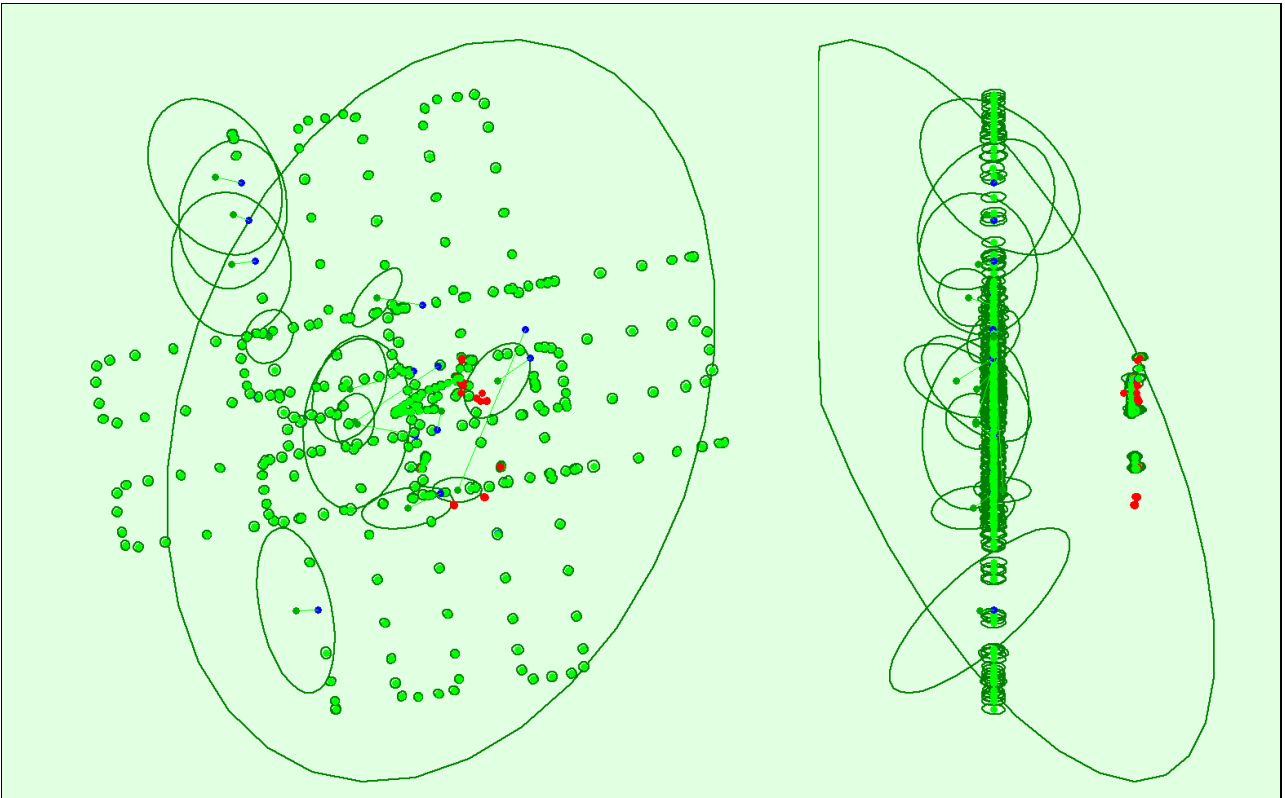
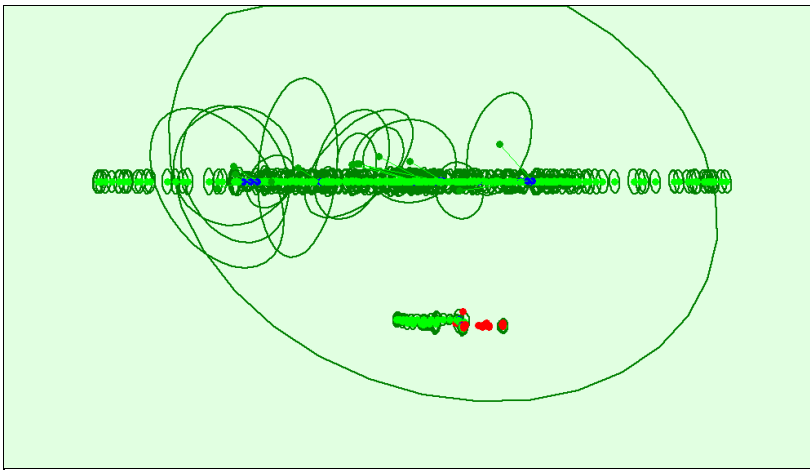


Figure 2: Top view of the initial image position. The green line follows the position of the images in time starting from the large blue dot.

? Computed Image/GCPs/Manual Tie Points Positions





Uncertainty ellipses 1000x magnified

Figure 3: Offset between initial (blue dots) and computed (green dots) image positions as well as the offset between the GCPs initial positions (blue crosses) and their computed positions (green crosses) in the top-view (XY plane), front-view (XZ plane), and side-view (YZ plane). Red dots indicate disabled or uncalibrated images. Dark green ellipses indicate the absolute position uncertainty of the bundle block adjustment result.

? Absolute camera position and orientation uncertainties



	X[m]	Y[m]	Z[m]	Omega [degree]	Phi [degree]	Kappa [degree]	Camera Displacement X[m]	Camera Displacement Y[m]	Camera Displacement Z[m]
Mean	0.002	0.002	0.004	0.023	0.009	0.021	0.007	0.007	0.008
Sigma	0.005	0.007	0.005	0.066	0.014	0.142	0.007	0.006	0.007

? Overlap

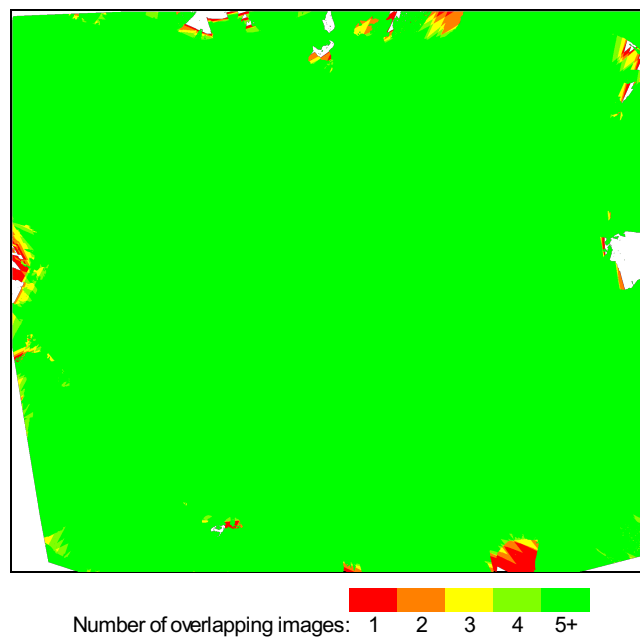


Figure 4: Number of overlapping images computed for each pixel of the orthomosaic. Red and yellow areas indicate low overlap for which poor results may be generated. Green areas indicate an overlap of over 5 images for every pixel. Good quality results will be generated as long as the number of keypoint matches is also sufficient for these areas (see Figure 5 for keypoint matches).

Bundle Block Adjustment Details



Number of 2D Keypoint Observations for Bundle Block Adjustment	5119265
Number of 3D Points for Bundle Block Adjustment	2020097
Mean Reprojection Error [pixels]	0.183

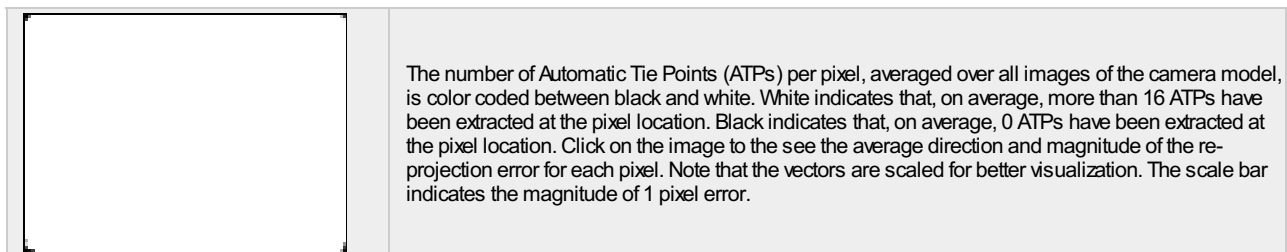
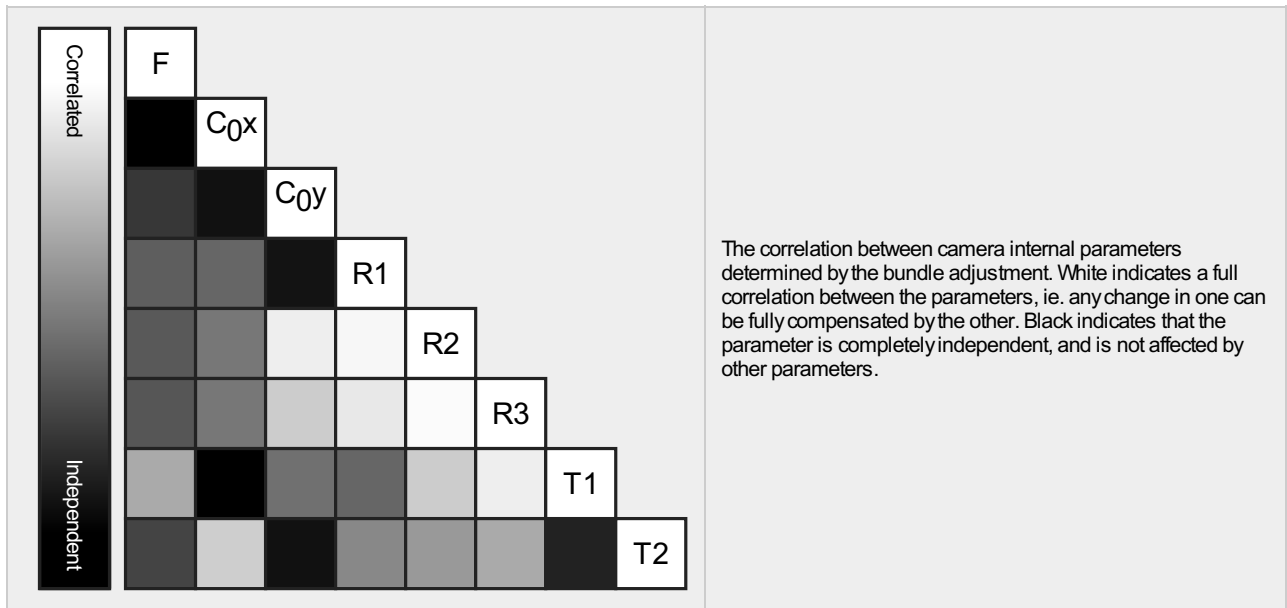
? Internal Camera Parameters

 **M4E_12.3_5280x3956 (1581F7FVC261F00DN1E3) (RGB). Sensor Dimensions: 17.923 [mm] x 13.429 [mm]**



EXIF ID: M4E_12.3_5280x3956

	Focal Length	Principal Point x	Principal Point y	R1	R2	R3	T1	T2
Initial Values	3620.570 [pixel] 12.290 [mm]	2640.000 [pixel] 8.961 [mm]	1978.000 [pixel] 6.714 [mm]	0.000	0.000	0.000	0.000	0.000
Optimized Values	3713.974 [pixel] 12.607 [mm]	2657.633 [pixel] 9.021 [mm]	1979.007 [pixel] 6.718 [mm]	-0.107	0.003	-0.020	-0.000	-0.001
Uncertainties (Sigma)	0.140 [pixel] 0.000 [mm]	0.153 [pixel] 0.001 [mm]	0.317 [pixel] 0.001 [mm]	0.000	0.000	0.000	0.000	0.000



? 2D Keypoints Table



	Number of 2D Keypoints per Image	Number of Matched 2D Keypoints per Image
Median	76571	11458
Mn	43907	145
Max	82100	58630
Mean	75886	13507

? 3D Points from 2D Keypoint Matches



	Number of 3D Points Observed
In 2 Images	1481971
In 3 Images	323362
In 4 Images	107253
In 5 Images	44998
In 6 Images	22976
In 7 Images	13203
In 8 Images	8096
In 9 Images	4969

In 10 Images	3362
In 11 Images	2143
In 12 Images	1635
In 13 Images	1267
In 14 Images	903
In 15 Images	701
In 16 Images	533
In 17 Images	430
In 18 Images	341
In 19 Images	278
In 20 Images	230
In 21 Images	178
In 22 Images	160
In 23 Images	148
In 24 Images	108
In 25 Images	106
In 26 Images	82
In 27 Images	69
In 28 Images	57
In 29 Images	63
In 30 Images	36
In 31 Images	41
In 32 Images	23
In 33 Images	35
In 34 Images	27
In 35 Images	34
In 36 Images	26
In 37 Images	17
In 38 Images	13
In 39 Images	14
In 40 Images	13
In 41 Images	21
In 42 Images	10
In 43 Images	15
In 44 Images	5
In 45 Images	7
In 46 Images	11
In 47 Images	7
In 48 Images	6
In 49 Images	7
In 50 Images	8
In 51 Images	2
In 52 Images	7
In 53 Images	11
In 54 Images	5
In 55 Images	6
In 56 Images	9
In 57 Images	5
In 58 Images	1
In 59 Images	4
In 60 Images	5
In 61 Images	4
In 62 Images	3
In 63 Images	2
In 65 Images	6
In 66 Images	7
In 67 Images	3
In 68 Images	4
In 69 Images	3

In 70 Images	1
In 73 Images	1
In 74 Images	1
In 75 Images	1
In 76 Images	2
In 78 Images	2
In 79 Images	3
In 89 Images	1

? 2D Keypoint Matches

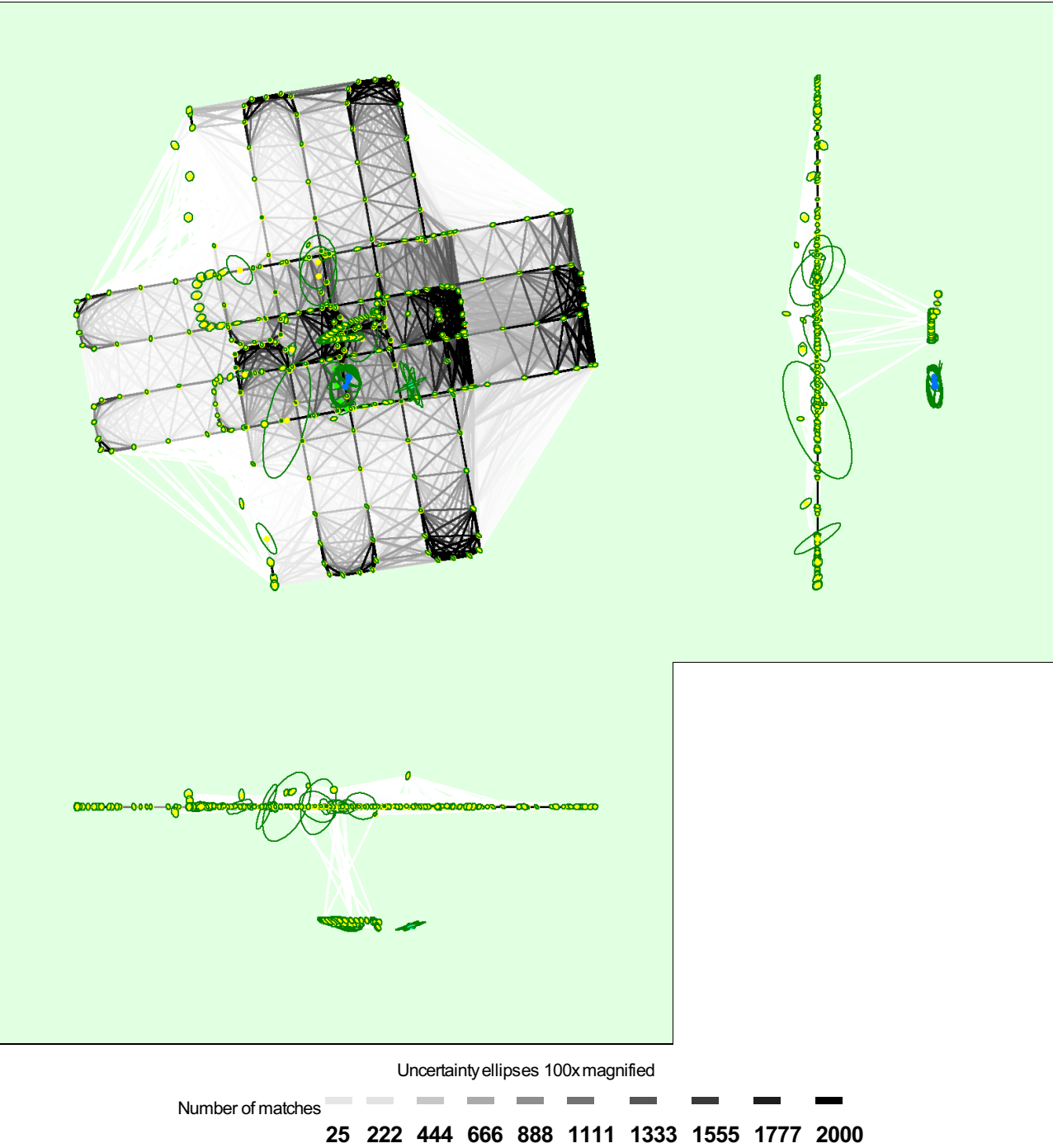


Figure 5: Computed image positions with links between matched images. The darkness of the links indicates the number of matched 2D keypoints between the images. Bright links indicate weak links and require manual tie points or more images. Dark green ellipses indicate the relative camera position uncertainty of the bundle block adjustment result.

? Relative camera position and orientation uncertainties

	X[m]	Y[m]	Z[m]	Omega [degree]	Phi [degree]	Kappa [degree]	Camera Displacement X[m]	Camera Displacement Y[m]	Camera Displacement Z[m]
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Mean	0.017	0.016	0.014	0.037	0.017	0.032	0.007	0.007	0.010
Sigma	0.015	0.021	0.012	0.104	0.030	0.217	0.008	0.009	0.012

Geolocation Details

Absolute Geolocation Variance

Mn Error [m]	Max Error [m]	Geolocation Error X[%]	Geolocation Error Y[%]	Geolocation Error Z[%]
-	-0.01	0.00	0.00	4.64
-0.01	-0.01	0.27	0.27	2.73
-0.01	-0.01	0.27	0.00	3.83
-0.01	-0.00	1.09	0.55	9.56
-0.00	-0.00	3.55	4.64	13.93
-0.00	0.00	44.81	43.44	16.39
0.00	0.00	45.08	46.17	16.12
0.00	0.00	3.01	4.37	10.93
0.00	0.01	1.64	0.27	9.02
0.01	0.01	0.27	0.27	4.10
0.01	0.01	0.00	0.00	3.28
0.01	-	0.00	0.00	5.46
Mean [m]		-0.000018	-0.000013	-0.000017
Sigma [m]		0.001780	0.001654	0.008235
RMS Error [m]		0.001780	0.001654	0.008235

Min Error and Max Error represent geolocation error intervals between -1.5 and 1.5 times the maximum accuracy of all the images. Columns X, Y, Z show the percentage of images with geolocation errors within the predefined error intervals. The geolocation error is the difference between the initial and computed image positions. Note that the image geolocation errors do not correspond to the accuracy of the observed 3D points.

Relative Geolocation Variance

Relative Geolocation Error	Images X[%]	Images Y[%]	Images Z[%]
[-1.00, 1.00]	88.80	89.07	73.22
[-2.00, 2.00]	96.17	98.63	92.08
[-3.00, 3.00]	99.18	99.18	96.45
Mean of Geolocation Accuracy [m]	0.002335	0.002335	0.006737
Sigma of Geolocation Accuracy [m]	0.000090	0.000090	0.000386

Images X, Y, Z represent the percentage of images with a relative geolocation error in X, Y, Z.

Geolocation Orientational Variance	RMS [degree]
Omega	80.071
Phi	1.640
Kappa	63.002

Geolocation RMS error of the orientation angles given by the difference between the initial and computed image orientation angles.

Rolling Shutter Statistics

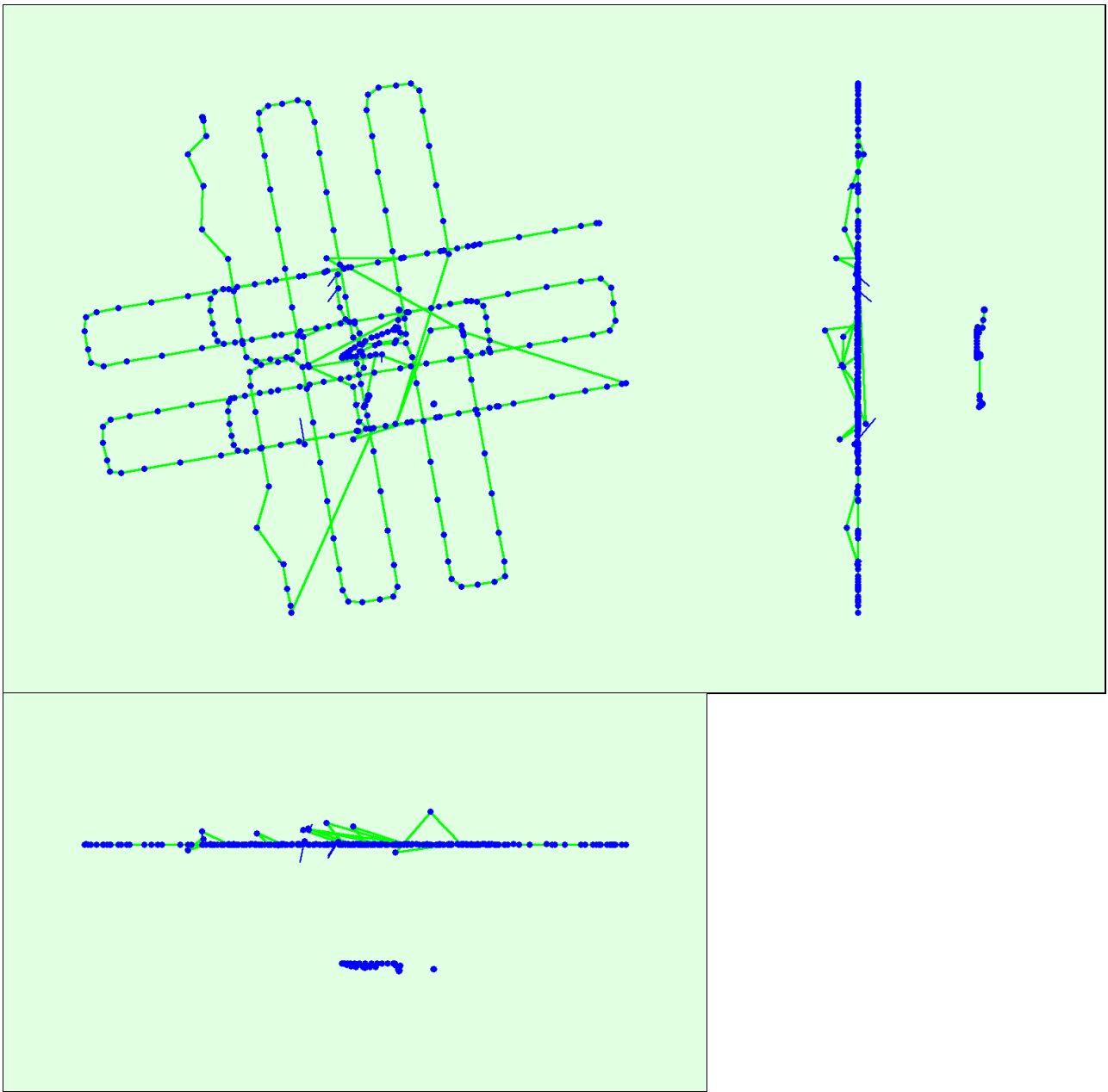


Figure 6: Camera movement estimated by the rolling shutter camera model. The green line follows the computed image positions. The blue dots represent the camera position at the start of the exposure. The blue lines represent the camera motion during the rolling shutter readout, re-scaled by a project dependant scaling factor for better visibility.

Median Camera Speed	6.1346 [m/s]
Median Camera Displacement During Sensor Readout)	0.055 [m]
Median Rolling Shutter Readout Time	12.0957 [ms]

Initial Processing Details

System Information

Hardware	CPU: AMD Ryzen 9 5900XT 16-Core Processor RAM: 64GB GPU: NVIDIA GeForce RTX 5070 (Driver: 32.0.15.9579)
Operating System	Windows 10 Pro, 64-bit

Coordinate Systems

Image Coordinate System	WGS 84
Output Coordinate System	WGS 84 / UTM zone 14N

Processing Options



Detected Template	3D Maps
Keypoints Image Scale	Full, Image Scale: 1
Advanced: Matching Image Pairs	Aerial Grid or Corridor
Advanced: Matching Strategy	Use Geometrically Verified Matching: no
Advanced: Keypoint Extraction	Targeted Number of Keypoints: Automatic
Advanced: Calibration	Calibration Method: Standard Internal Parameters Optimization: All External Parameters Optimization: All Rematch: Auto, yes

Point Cloud Densification details



Processing Options



Image Scale	multiscale, 1/2 (Half image size, Default)
Point Density	Optimal
Minimum Number of Matches	3
3D Textured Mesh Generation	yes
3D Textured Mesh Settings:	Resolution: Medium Resolution (default) Color Balancing: yes
LOD	Generated: no
Advanced: 3D Textured Mesh Settings	Sample Density Divider: 1
Advanced: Image Groups	group1
Advanced: Use Processing Area	yes
Advanced: Use Annotations	yes
Time for Point Cloud Densification	01h:57m:13s
Time for Point Cloud Classification	50s
Time for 3D Textured Mesh Generation	15m:32s

Results



Number of Generated Tiles	1
Number of 3D Densified Points	25131914
Average Density (per m ³)	399.96

DSM, Orthomosaic and Index Details



Processing Options



DSM and Orthomosaic Resolution	1 x GSD (1.76 [cm/pixel])
DSM Filters	Noise Filtering: yes Surface Smoothing: yes, Type: Sharp
Raster DSM	Generated: yes Method: Inverse Distance Weighting Merge Tiles: yes
Orthomosaic	Generated: yes Merge Tiles: yes GeoTIFF Without Transparency: no Google Maps Tiles and KML: yes
Grid DSM	Generated: yes, Spacing [cm]: 100
Raster DTM	Generated: yes Merge Tiles: yes
DTM Resolution	5 x GSD (1.76 [cm/pixel])

Contour Lines Generation	Generated: yes Contour Base [m]: 0 Elevation Interval [m]: 0.5 Resolution [cm]: 300 Minimum Line Size [vertices]: 100
Time for DSM Generation	08m:22s
Time for Orthomosaic Generation	56m:21s
Time for DTM Generation	02m:16s
Time for Contour Lines Generation	01s
Time for Reflectance Map Generation	00s
Time for Index Map Generation	00s